EXAMPLE: ISING MODEL

- Let $\mathbb{X} = \{-1,1\}^V$ where V is the a d-dimensional lattice
- The energy forces neighbouring spins to prefer alignment, penalised by an external field

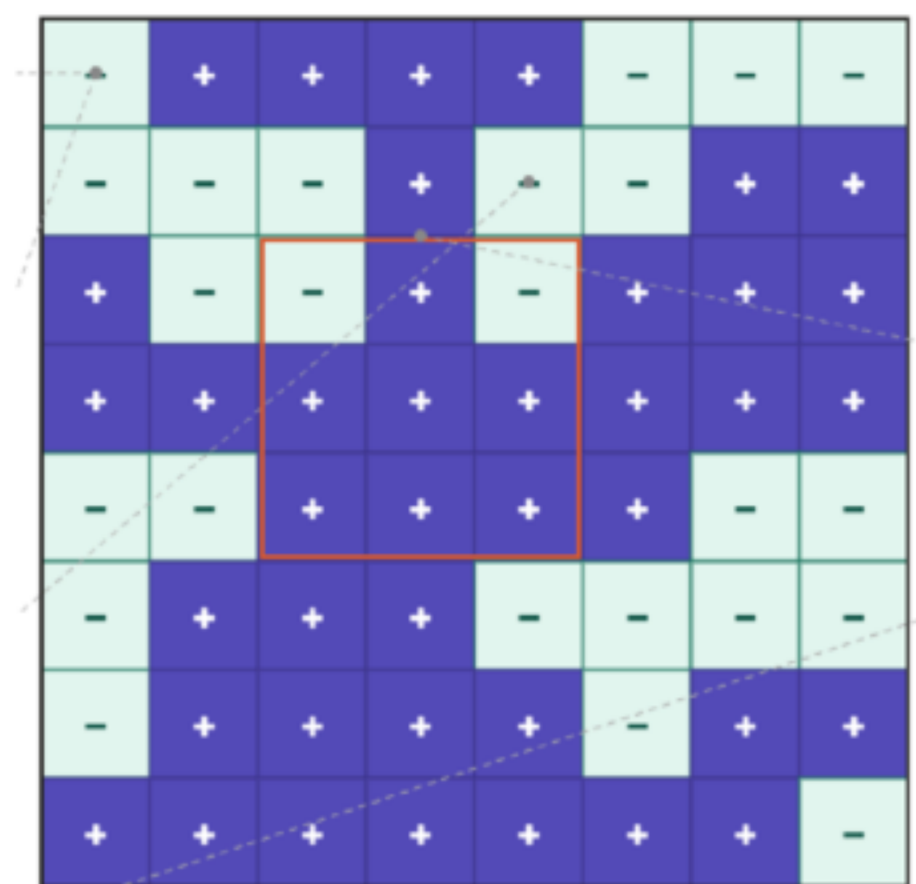
$$U(x) = -J\sum_{i\sim j} x_i x_j - h\sum_i x_i$$

  - The first term favours aligned neighbours with coupling parameter J
  - The second term favours all spins aligning with magnetic moment h

$$\pi_\beta(x) \propto \exp\left(\beta J\sum_{i\sim j} x_i x_j + \beta h\sum_i x_i\right)$$

  - Total magnetisation of the state x is $m(x)$

$$m(x) = \sum_{i=1} x_i$$



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